



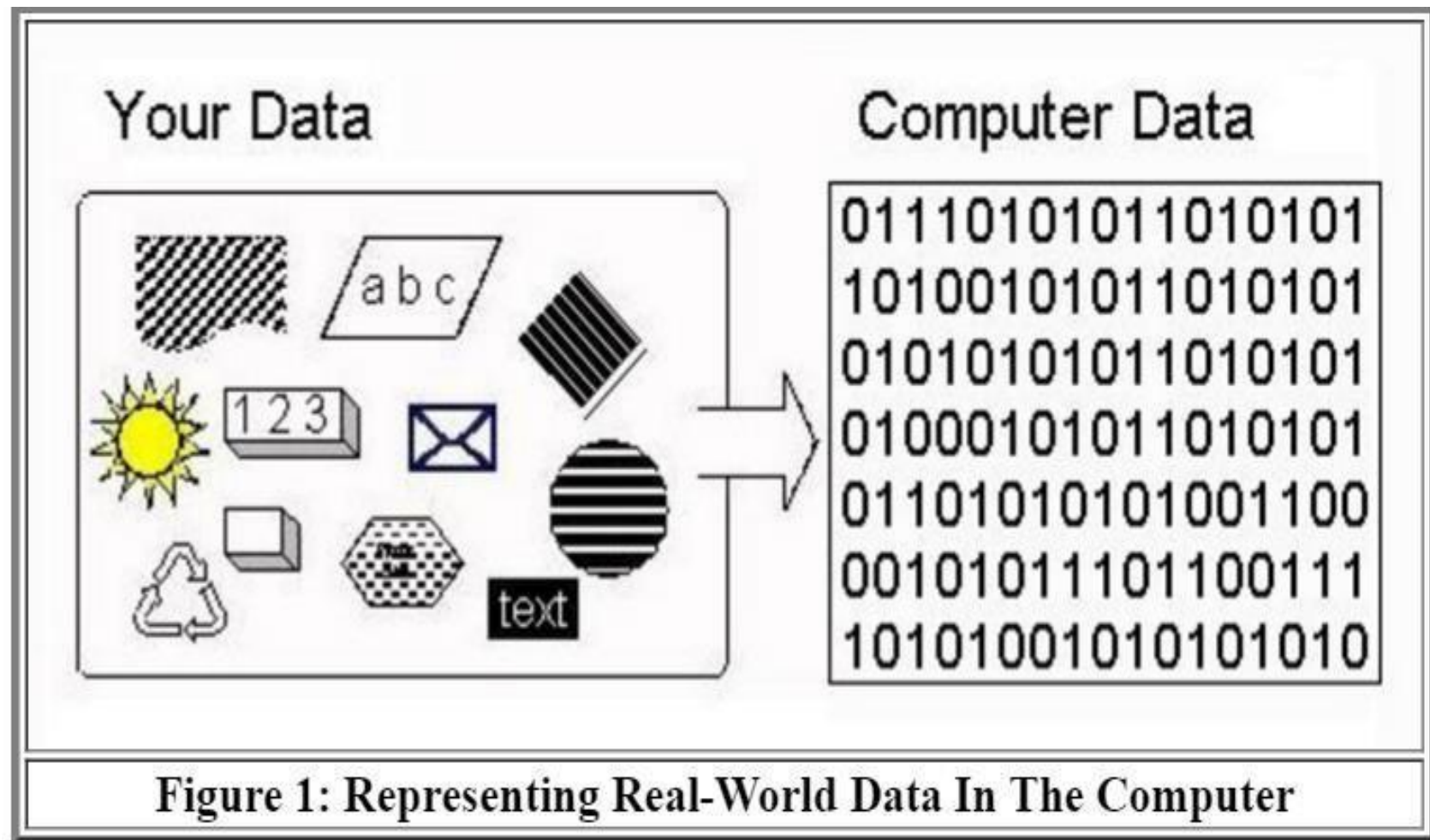
ICT 1201

Fundamentals of Computer
Systems

Information Representation in
Computer

Representing Data

- ❑ Computers and other digital devices work with text ,numbers ,music ,images ,speech and video.
- ❑ The amazing aspect of digital technology is that it distills all these different elements down to simple pulses of electricity and stores them as 0s and 1s.



Representing Data

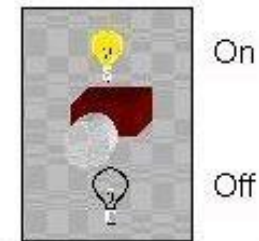
- ❑ Computers are Electronic Machines.
- ❑ The processor has got billions of little electronic switches inside it.
- ❑ The computer uses electricity, not mechanical parts, for its data processing and storage.
- ❑ Electricity can flow through switches; if the switch is closed, the electricity flows; if the switch is open, the electricity does not flow.
- ❑ To process real world data in the computer, we need a way to represent the data in switch.
- ❑ Computers do this representation using a binary coding system.

Binary and Switches

- ❑ Binary is a mathematical number system : a way of counting.
- ❑ The computer has switches to represent data and switches have only two states: ON and OFF.
- ❑ Binary has two digits to do the counting: 0 and 1 – a natural fit to the two states of switch (0=OFF, 1=ON)

Bits and Bytes

- ❑ One binary digit (0 or 1) is referred as a bit, which is short for *binary digit*.
- ❑ One bit can be implemented by one switch.

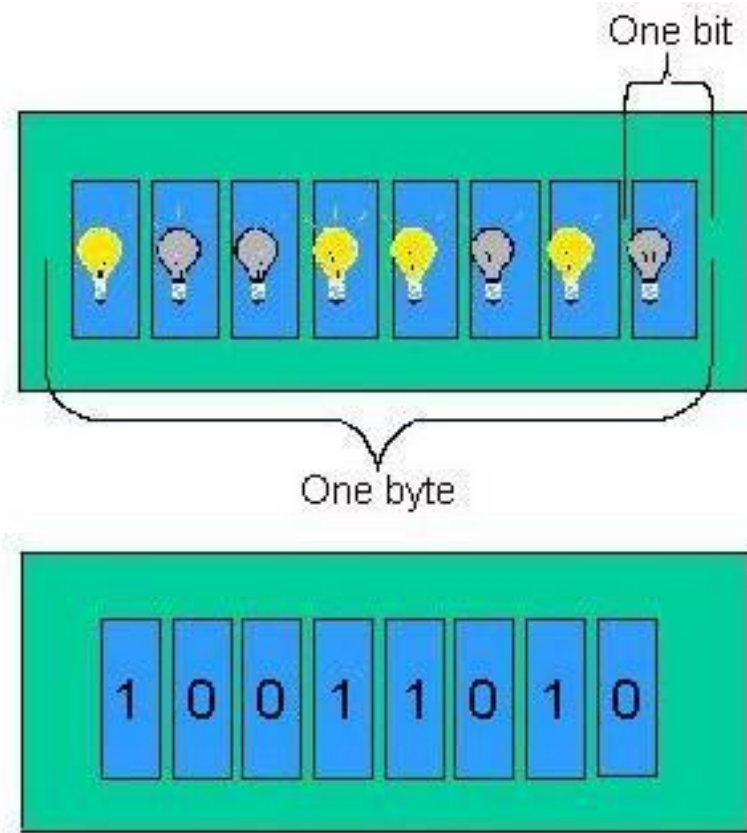


One bit can be represented as a switch that is either on or off.

- ❑ Bits can be grouped together into large chunks to represent data.

0	1 bit
1	1 bit
0110	4 bits
01101011	8 bits

- ❑ Computer designers use eight bit chunks called bytes as the basic unit of data. A byte is implemented with eight switches as follows.



- A single byte can represent many different kinds of data. What data it actually represents depends on how the computer uses the byte.
- As an example, the byte:



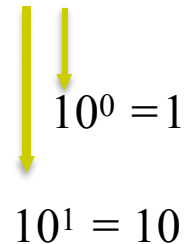
01000011

- Can represent the integer 67
- The character “C”
- The 67th level of darkness for a dot in a picture
- An instruction to the computer like “move to memory”

Integers

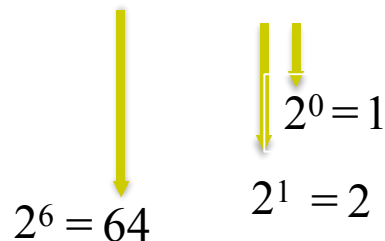
- ❑ Integer numbers are represented by counting in binary.
- ❑ What “places” we use depends on the counting system.
- ❑ In our decimal system, which we call *Base 10*, we use powers of 10. Ten to the zero power is 1, so the counting starts in the "one's place". Ten to the one power is 10, so the counting continues in the "ten's place". Ten to the second power (10 squared) is 100, so we continue in the "hundred's place". And so on.

- ❑ Binary is *Base 2*. Thus, the "places" are two to the zero power ("one's place"), two to the one power ("two's place"), two to the second power ("four's place"), two to the third power ("eight's place"), and so on.
- ❑ Ex: Decimal Integer 67



$10^0 = 1$
 $10^1 = 10$

Same decimal number can be represent in binary format as follows ;



$2^6 = 64$
 $2^0 = 1$
 $2^1 = 2$

- ❑ The largest decimal integer you can store in one byte is 255.
- ❑ So computer use several bytes together to store large integers.

Numbers, as known in the decimal-system	Same numbers in binary system
0	0
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000

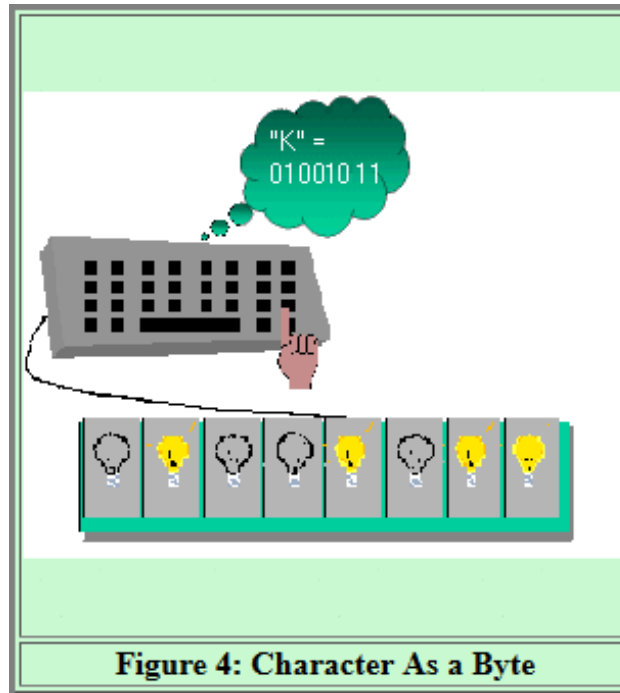
Characters

- ❑ The computer uses a single byte to represent a single character.
- ❑ There must be agreement on which groups of bits represent which characters.
- ❑ The code called ASCII, which stands for American Standard Code for Information Interchange, uses 7 bits for each character.
- ❑ A more common version is ASCII-8, also called extended ASCII, which uses 8 bit per character and can represent 256 different characters.
- ❑ Ex: the letter A is represented by ASCII code 65 is equivalent to binary code 01000001

Character	Bit pattern	Byte number		Character	Bit pattern	Byte number
A	01000001	65		¼	10111100	188
B	01000010	66		.	00101110	46
C	01000011	67		:	00111010	58
a	01100001	97		\$	00100100	36
b	01100010	98		\	01011100	92
o	01101111	111		~	01111110	126
p	01110000	112		1	00110001	49
q	01110001	113		2	00110010	50
r	01110010	114		9	00111001	57
x	01111000	120		©	10101001	169
y	01111001	121		>	00111110	62
z	01111010	122		‰	10001001	137

Part of ASCII-8 Code

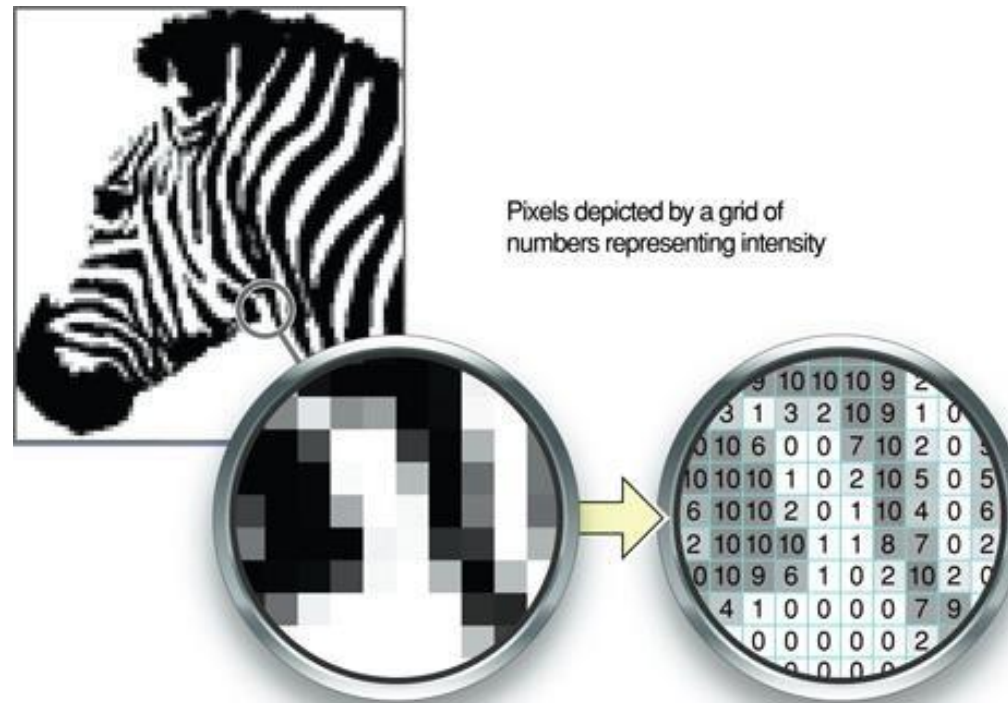
- What will happen when you type “K”



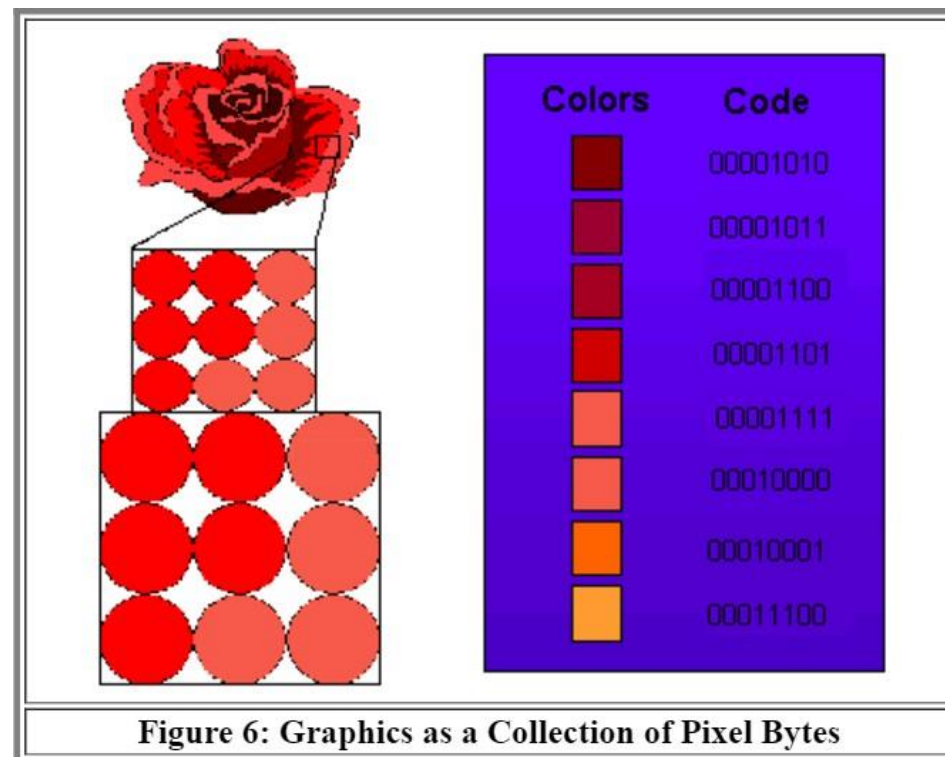
- Circuitry on the keyboard and in the computer converts the “K” to the byte 01001011 and store the letter in the computer’s memory as well as instructing the monitor to display it.

Picture and Graphic Data

- ❑ A grid of pixels represents computer graphic data like pictures, frames of a movie, drawings or frames of an animation.



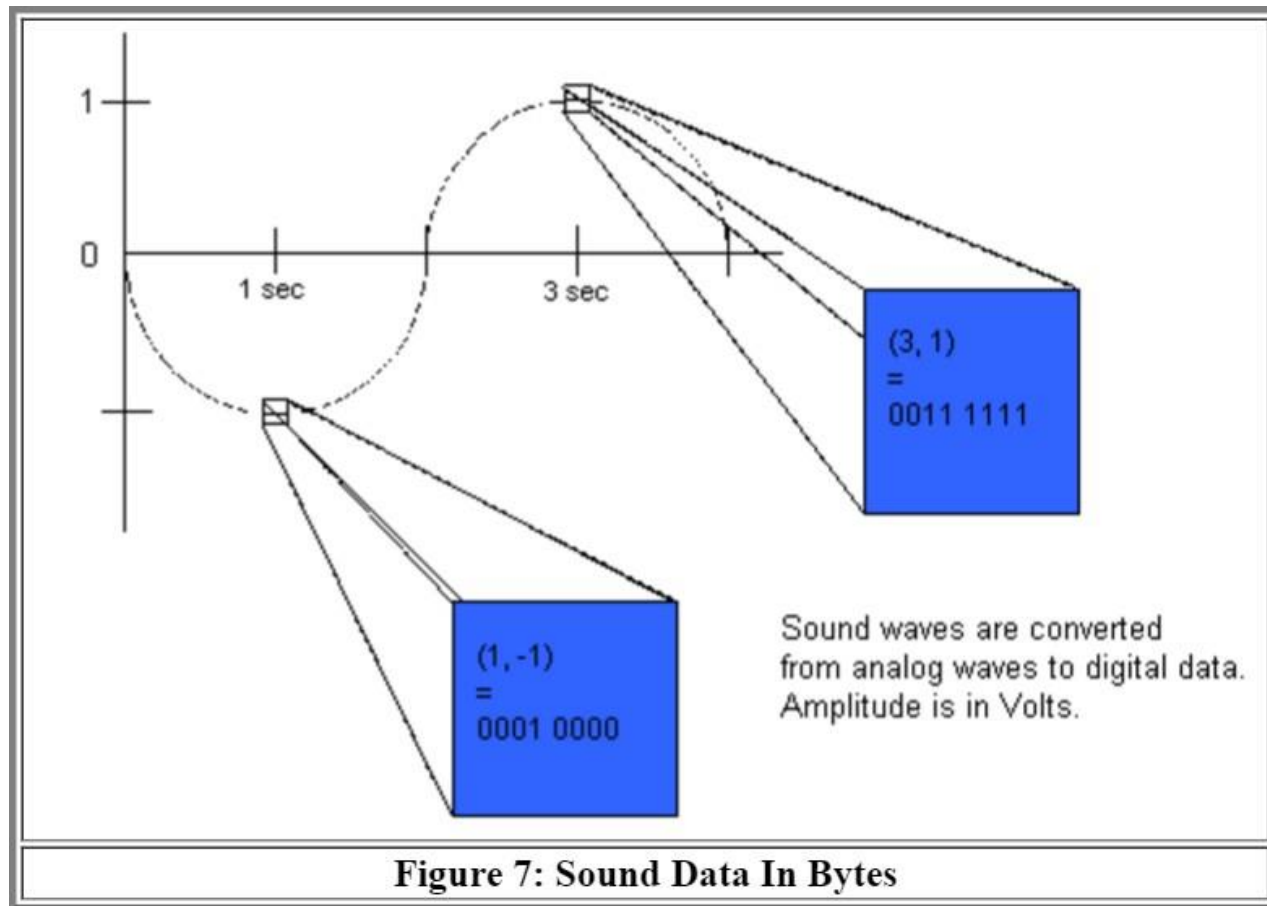
- ❑ “Pixel” is short for *picture element*.
- ❑ In simple graphics, a byte can represent a single pixel. Modern video games and colorful graphics use several bytes for each pixel.



- Different image formats are available.
 - Bitmap
 - Stores every pixel of the image.
 - JPEG
 - Stores the patterns of pixels.
 - Compressed formats.
 - GIF
 - Stores the patterns of pixels.
 - Compressed formats.

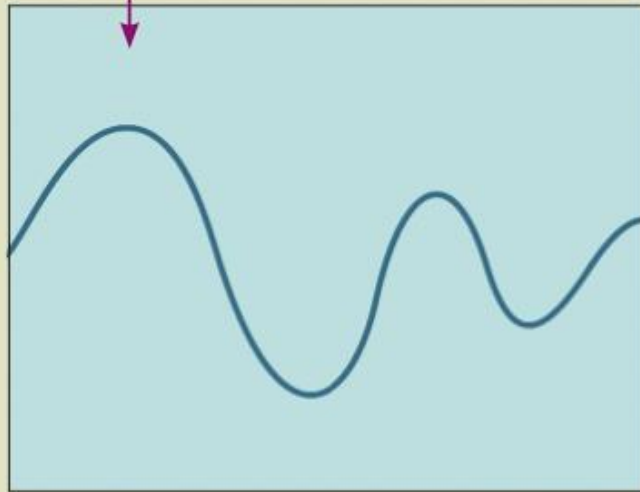
Sound

- ❑ Sound occurs naturally as an analog wave

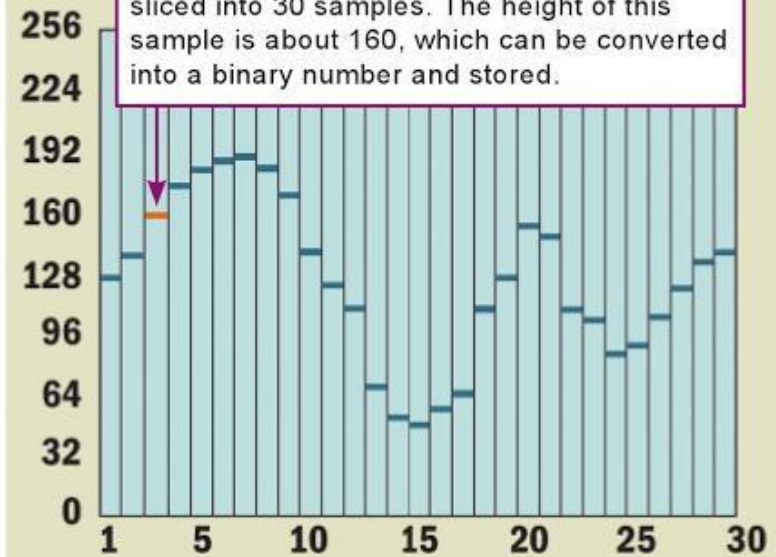


- ❑ The process of taking analog data and making it digital is called analog to digital conversion.
- ❑ To convert an analog wave into digital, converters use a process called sampling.
- ❑ Sample the height of the sound wave at regular intervals of time.
- ❑ If one byte is used to hold a single sample of an analog wave, then the wave can be one of 256 different heights (0 being the lowest height and 255 being the highest).

An analog sound wave is a smooth curve of continuous values.



To digitize a wave, it is sliced into vertical segments, called samples. For purposes of illustration, this one-second sound wave was sliced into 30 samples. The height of this sample is about 160, which can be converted into a binary number and stored.



- Like ASCII for characters and GIF, JPEG for images, sound has several agreed-upon formats as follows for representing samples in bytes.
 - WAV
 - Every sample is stored.
 - MP3
 - Compressed format — does not store every sample.
 - Store only samples that the human ear can hear and then condenses these samples to patterns.

Program Data

- ❑ The programming process allows human to write instructions in an English like way.
- ❑ A software program called a compiler then transform the English like text into the bytes for instructions that the computer understands.
- ❑ Like all other kinds of data, there are agreed upon formats for computer instructions too.

```
INPUT A, B
C = A + B
IF C > 100 THEN GOTO LABEL1
IF C < 100 THEN PRINT "You
    Lose!!"; END
PRINT "This is a winner!"
LABEL1: D = 100 - C
PRINT "You were" D "over 100."
END
```



Compiler
Program



```
INPUT    11001011
=        00001000
>        00001001
<        00001010
GOTO     10001010
PRINT    11001111
END      10001111
```



To Processor

How Does the computer know what a byte represents ?

- ❑ From the *context* in which the computer uses the byte.
- ❑ If it sends the byte to a speaker, the 67th level of sound is produced.
- ❑ If it sends the byte to a monitor or printer, a pixel with the 67th level of darkness is produced.
- ❑ More accurately, if the byte were coded with a standard coding technique, like ASCII for characters, GIF for pictures, and WAV for sounds, then when the computer sends the byte to a device, the data corresponding to that coding is produced by the device.

THANK YOU